Industrial Internship Report on

”Expense Tracker”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was "Console-Based Expense Tracker Application" developed using Java. It helps users record, manage, update, and track their daily expenses, organize them into categories, and generate expense reports using a simple console-based interface. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

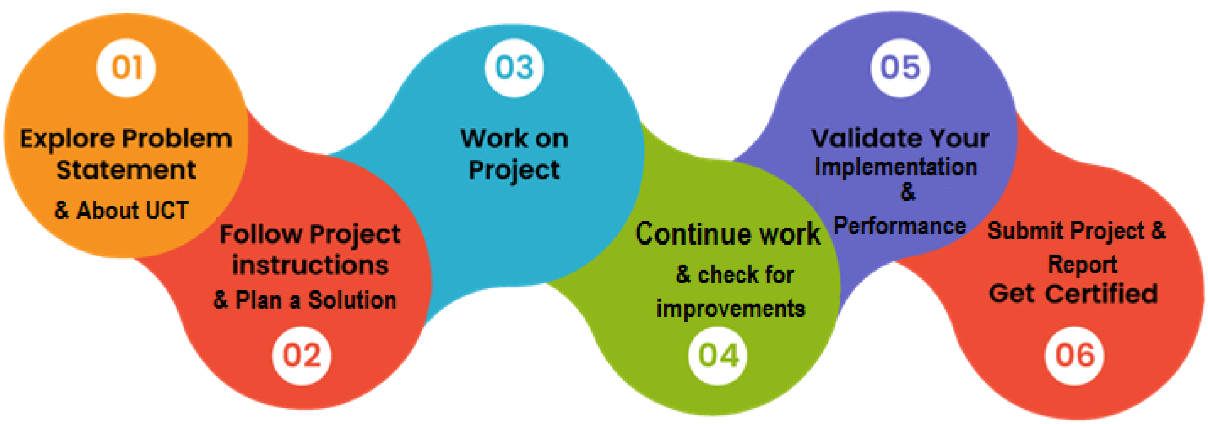
During the six-week internship, I learned Core Java fundamentals, Full-Stack Web Development basics, and developed a **Console-Based Expense Tracker Application**. The internship enhanced my programming, problem-solving, and project development skills.

An internship provides practical industry experience and helps bridge the gap between academic knowledge and real-world applications. It also improves technical and professional skills.

I developed a **Console-Based Expense Tracker Application** using Java. It allows users to manage expenses, organize categories, and generate reports through a simple console interface.

Upskill Campus and UCT provided an opportunity to learn industry-relevant technologies, complete a real-world project, and improve practical software development skills.

The internship followed a structured plan that included learning Java concepts, completing quizzes, developing the project, testing the application, and preparing the final report.



I improved my knowledge of Core Java, object-oriented programming, file handling, and problem-solving. Overall, it was a valuable learning experience.

I sincerely thank **Upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd. (UCT)**, my mentors, faculty members, friends, and family for their continuous support and guidance..

Take every internship seriously, practice regularly, and work on real-world projects. Continuous learning and hands-on experience are the keys to becoming a successful software developer.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

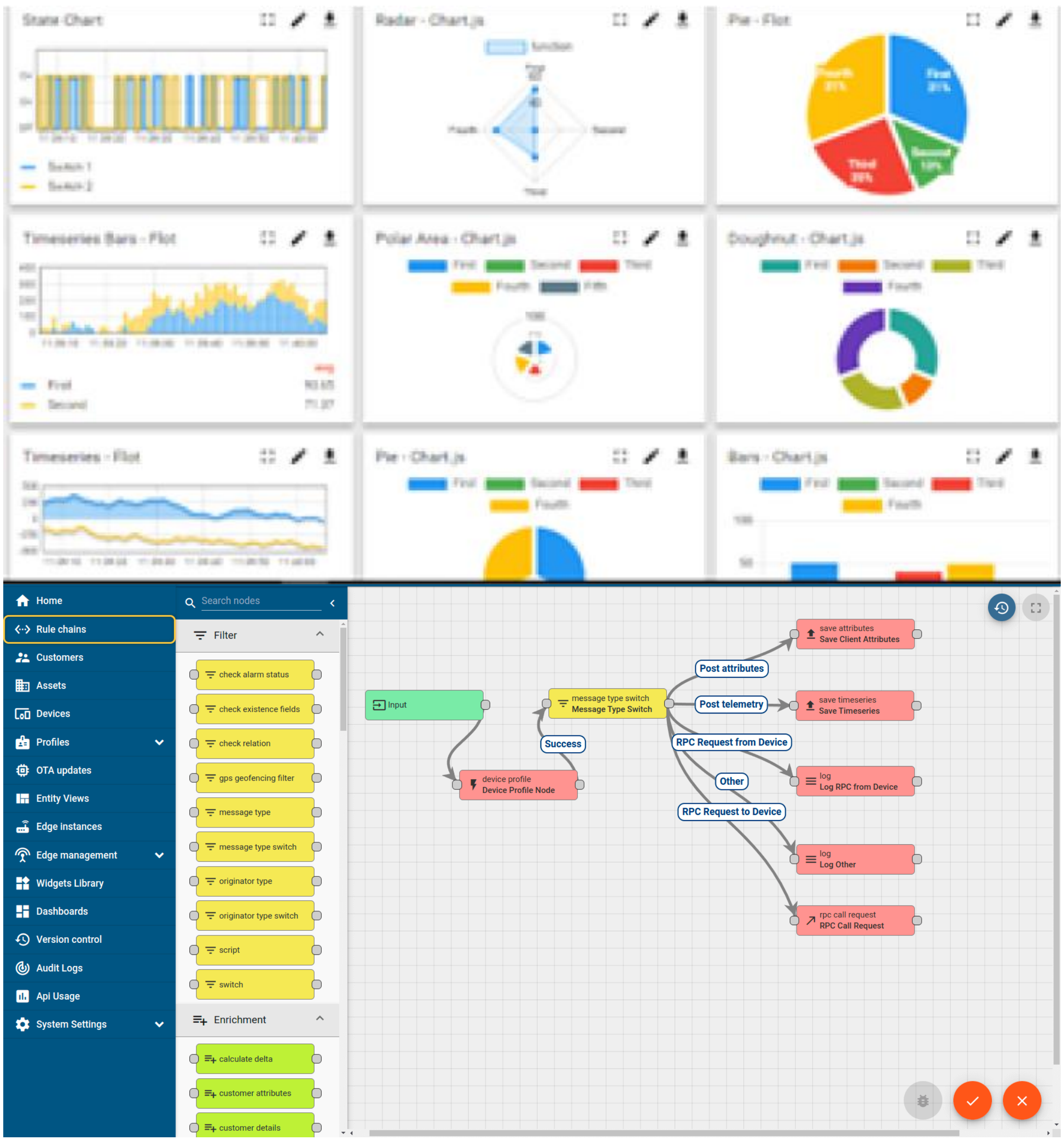
i. UCT IoT Platform(uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

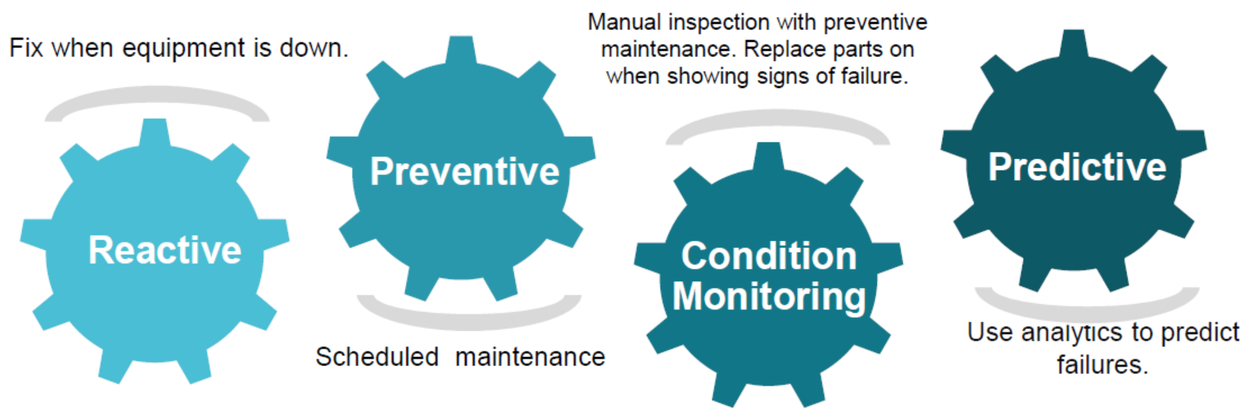
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

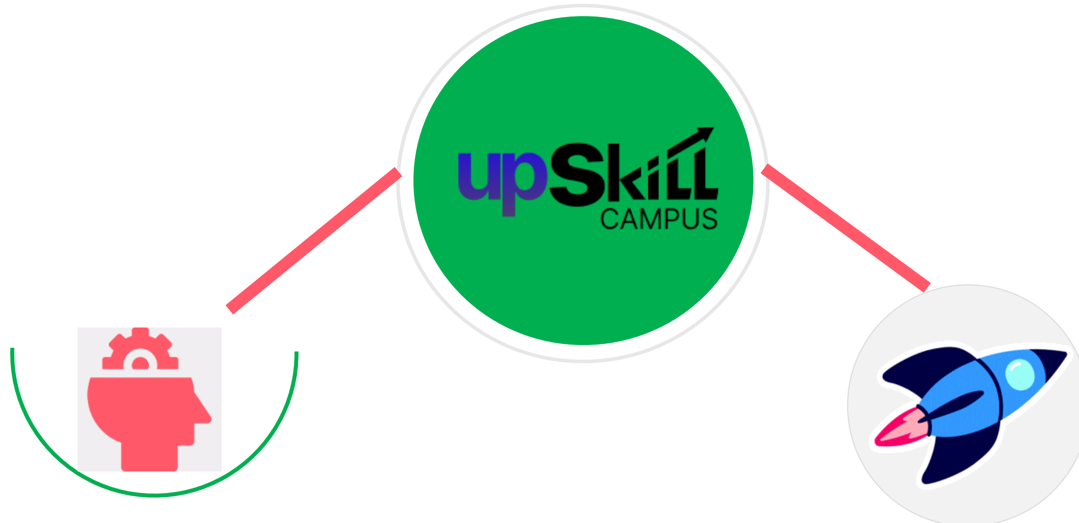
UCT isproviding Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

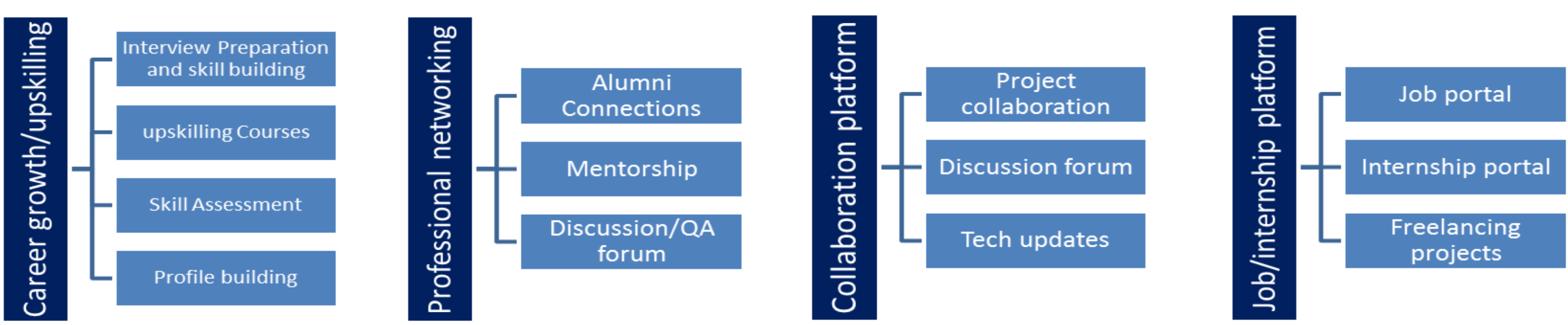
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛to have Improved understanding of our field and its applications.

☛to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Upskill Campus Learning Materials

[2] The IoT Academy Learning Resources

[3] Oracle Java Documentation

2.6 Glossary

Terms	Acronym
Java Development Kit	JDK
Java Runtime Environment	JRE
Java Virtual Machine	JVM
Object-Oriented	OOP

HyperText Markup Language	HTML

3 Problem Statement

The assigned problem statement was to develop a **Console-Based Expense Tracker Application** using Core Java. The application is designed to help users manage their daily expenses efficiently by allowing them to record, update, delete, and view expense details. Users can categorize their expenses, filter records based on different criteria, and generate reports to analyze their spending patterns. The application also uses file handling to store data permanently, ensuring that expense records remain available even after restarting the program.

Key Features:

- Add, update, and delete expenses.
- Manage expense categories.
- Search and filter expense records.
- Generate monthly and category-wise reports.
- Store data using file handling.
- Provide a simple and user-friendly console interface.

This project demonstrates the practical application of Core Java concepts such as Object-Oriented Programming, Collections, Exception Handling, File Handling, and Input Validation to solve a real-world personal finance management problem.

4 Existing and Proposed solution

4.1.1 Existing Solution

Many people manage their daily expenses using notebooks, spreadsheets, or mobile applications. Manual methods are time-consuming and prone to errors, while many existing applications are complex or require internet access and user accounts.

4.1.2 Limitations of Existing Solutions

- Manual expense tracking is difficult to maintain.
- Chances of calculation and data entry errors.
- Limited expense analysis in traditional methods.
- Many applications require internet connectivity.
- Some applications have unnecessary features for basic users.

4.1.3 Proposed Solution

The proposed solution is a **Console-Based Expense Tracker Application** developed using Core Java. It allows users to add, update, delete, search, and filter expenses. Users can also manage expense categories, generate reports, and store data permanently using file handling. The application is simple, lightweight, and easy to use.

4.1.4 Value Addition

- User-friendly console interface.
- Automatic expense report generation.
- Category-wise expense management.
- Permanent data storage using file handling.
- Fast searching and filtering of expenses.
- Improved organization and tracking of personal finances.

4.2 Code submission (Github link)

https://github.com/akil9979/upskillcampus.git

4.3 Report submission (Github link) :

5 Proposed Design/ Model

The proposed system follows a modular design where the user interacts with a console-based menu to perform various expense management operations. The system validates user inputs, processes the requested operation, stores data using file handling, and displays the results. This modular approach improves maintainability, scalability, and usability.

5.1 High Level Diagram (if applicable)

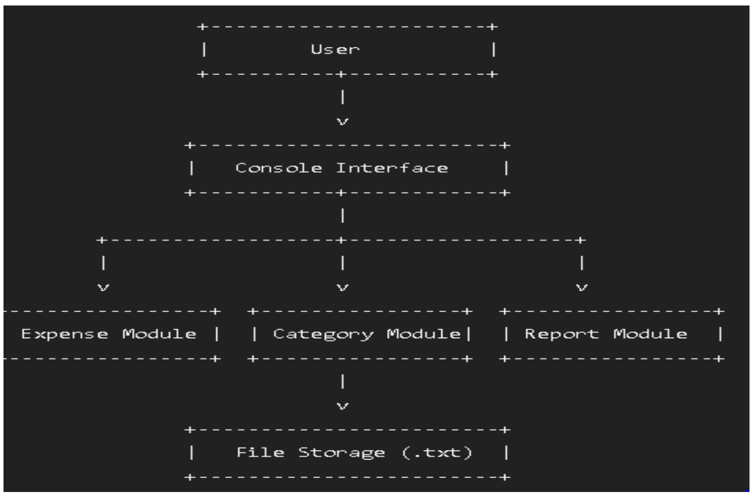
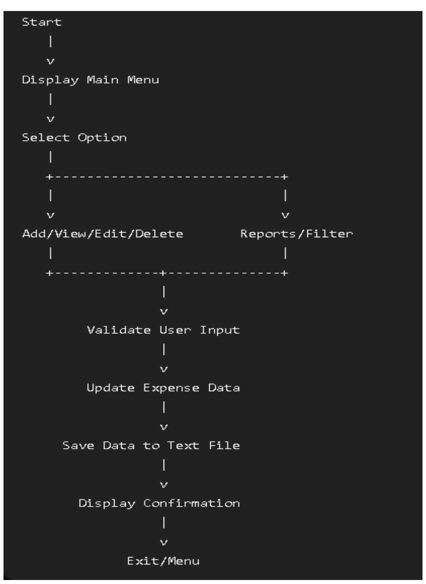


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The application uses a **menu-driven console interface** that allows users to:

- Add new expenses
- View all expenses
- Update existing expenses
- Delete expenses
- Search expenses
- Manage categories
- Filter expenses
- Generate reports
- Save and retrieve data using file handling

5.3.1 Technologies Used

- Java
- Object-Oriented Programming (OOP)
- Collections (`ArrayList`)
- File Handling
- Exception Handling
- Scanner Class
- LocalDate API

6 Performance Test

The performance of the Console-Based Expense Tracker Application was evaluated by testing its core functionalities under different user inputs. The application was designed to provide accurate expense management while maintaining fast response time and reliable data storage.

The main constraints considered during development were input validation, data accuracy, execution speed, memory usage, and persistent data storage using file handling. Appropriate exception handling and validation techniques were implemented to ensure smooth execution and prevent invalid user inputs.

The application successfully handled expense recording, updating, deleting, searching, filtering, and report generation without any major performance issues.

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Add Expense	Expense added successfully	Passed
Update Expense	Expense updated correctly	Passed
Delete Expense	Expense removed successfully	Passed
Search Expense	Correct expense displayed	Passed
Filter Expenses	Matching records displayed	Passed
Generate Report	Report generated correctly	Passed
Invalid Input	Error message displayed	Passed
Save & Reload Data	Data stored and loaded successfully	Passed

6.2 Test Procedure

- Started the application.
- Added multiple expense records.
- Updated and deleted existing records.
- Searched expenses using different criteria.
- Filtered expenses by category and amount.
- Generated expense reports.
- Restarted the application to verify file storage.
- Tested invalid inputs to verify exception handling.

6.3 Performance Outcome

The application performed successfully during testing. All major functionalities worked as expected with accurate results. Input validation prevented invalid data entry, file handling ensured permanent data storage, and the application responded quickly even with multiple expense records. Overall, the system met the project requirements and demonstrated reliable performance.

7 My learnings

During this internship, I gained valuable knowledge in **Core Java** and **Full-Stack Web Development**. I learned Java fundamentals, platform independence, object-oriented programming concepts, file handling, exception handling, and the basics of HTML and web development. I also improved my problem-solving, debugging, and logical thinking skills while developing the **Console-Based Expense Tracker Application**.

Working on a real-world project helped me understand the complete software development process, from analyzing requirements to designing, implementing, testing, and documenting the application. This internship also enhanced my confidence, communication skills, and ability to work independently.

Overall, this experience has strengthened my technical foundation and prepared me for future opportunities in software development. The knowledge and practical experience gained during this internship will help me build more advanced applications and contribute effectively in my professional career.

8 Future work scope

The current application provides basic expense management features through a console interface. In the future, it can be enhanced by adding a graphical user interface (GUI) or converting it into a web or mobile application for better user experience. A database such as MySQL or MongoDB can be integrated to improve data management and scalability. Additional features like user authentication, budget planning, expense charts, data export (PDF/Excel), cloud backup, and email notifications can also be implemented. These improvements will make the application more efficient, secure, and suitable for real-world use.